Preceding pages
The Chephren Pyramid at Giza with the Sphinx.

Mummies, Mummification Techniques and the Cult of the Dead in Ancient Egypt—A Chronological Overview

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Page 74
Detail of the outer coffin of Nes-pa-kai-chuti from the Lippisches Landesmuseum, Detmold.

The term "mummy" is understood in the natural sciences to be a preserved human or animal corpse, including skin, hair, and flesh, regardless of whether it has been created by natural processes or by artificial methods. In addition, in Egyptology, a wrapped body is also usually referred to as a mummy when the silhouette of the dead person has been preserved, yet only skeletal remains are found within the wrappings.

By "mummification", Egyptologists understand not only the method of carrying out the preservation of soft tissues of the body, but also its broader meaning of the ritual treatment of the dead person before his final burial, which includes embalming materials and bandages. The following article deals with the technical and religious aspects of body preservation.



1 Partially mummified skeleton of a woman, Hierakonpolis (HK 42, burial 85) c. 3500 BC.

Predynastic and Early Dynastic Egypt: Natural Mummies, Burial Customs and the Beginnings of a Cult of the Dead

Verifiable remains of settlements of prehistoric cultures in Egypt go back as far as the 6th millennium BC. The most common burial position of a body in prehistoric times was the flexed pose (fig. 1): the body lay, wrapped in matting, in a shallow grave, with burial objects placed next to it. At first, the body was laid with the head pointing west, as a rule, but over time an eastward orientation predominated. The civilizations based in Upper Egypt buried their dead in desert areas bordering villages, where the body was exposed to sun and salty desert sand, which led to preservation through desiccation. Today it is assumed that this effect caused by the environment made its mark on the religious concept of a death cult, which took physical preservation as a necessary condition for the afterlife.

With the formation of the first population centres and elite cemeteries, in c. 3700 BC, there was a change in grave structures. Grave pits became larger and their walls were partially sealed with bricks, which may have protected the bodies from sun and sand, but which also hindered natural mummification. For a long time, received opinion in Egyptology held that the decomposition observed from altered burial conditions was the only impetus for the development of artificial mummification. This would be because, once the establishment of a cult of the dead was fixed, a corpse had to be protected from decay at all costs. Following the latest finds in a workers' cemetery in Hierakonpolis (HK 43), dating from c. 3600 BC, however, it seems that other concepts of the afterlife were already playing a role: the bodies of

three prominent women of the time were found, covered with mats, with their necks and hands wrapped in fine linen soaked in resin—a measure which, compared to exposure to sun and sand, was not conducive to greater preservation of soft tissues. One of the women had had her throat cut after her death; another woman had an organ removed, which was then wrapped in a resin-soaked cloth and put back in her body (fig. 1). It is possible to speculate as to why this was done. It appears that the most important priority was not the long-term appearance of the body, but rather the preservation of particular body functions. Therefore, those parts of the body that were necessary for the consumption of food were together and could not be lost.

Linen soaked in resin has also been found in the graves of the oldest protohistoric burial sites, such as Mostagedda (c. 4000 BC). If the goal of *postmortem* manipulation was "mummification", in the sense of preservation of soft tissues, even if in the beginning it was only specific tissues, this can be seen as evidence of the beginnings of artificial preservation of bodies going back as far as 6,000 years. The goal itself, as shown by the finds, had not yet been reached.

The creation and development of a unified state in Egypt took place through a long process (0 Dynasty) around the year 3000 BC. Resin-soaked linen bandages, such as were known early in history, were found with skeletal remains in the necropolises of the 1st and 2nd dynasties (c. 3050–2700 BC) in Saqqara and Abydos. The better-preserved burials in Saqqara show bodies in flexed pose, wrapped in heavy layers of linen and buried in wooden coffins. Their soft tissues are also not preserved. The bodies found in the royal tombs of Abydos have only a few impregnated linen cloths with pieces of bone clinging to them. They come from the Dewen tomb area and were originally assigned to the royal graves of Djer, Djet, Dewen and Merneith. What is probably the best-known fragment was found by a member of Petrie's excavation team in 1901 in the tomb complex of King Djer (c. 3000 BC): a lower arm, wrapped in fine linen and wearing four bracelets made of gold, turquoise, lapis lazuli and amethyst. Preservation of the skin or muscle tissues had not occurred (fig. 2). It is generally accepted that the arm was that of Djer or his wife, Merneith. At the beginning of the 20th century no particular importance was attached to the find; the curator of the Cairo museum at the time is said to have thrown away the bandages along with the arm. The bracelets are still on display in the museum.



2 The linen-wrapped lower arm of King Djer or his consort Merneith, Abydos, c. 3000 BC.

The picture of how the dead were handled during the pre- and protohistoric eras is very incomplete. A few bodies (in El Adaima, Hierankonpolis, Naqada, but also later, during the Old Kingdom, in Giza, Meidum, and Deshasha) were dismembered, or had skin and muscle tissue removed from their bones, long before burial but certainly after death. Other bodies show the first signs of the application of mummification techniques. The enormous importance of the body's transitional period from time of death to burial is evident in the various forms of body treatment. A general explanation for the dismemberment of the bodies continues to be the subject of heated debate (Dougherty and Friedman 2008).

It is not until the following era, in the ritual texts of the dead, that a written description of the concepts connected to dismemberment exist, such as this excerpt from a text from the pyramid of Teti (PT 413; §§ 735b–736c): "Awake, O King, raise yourself, receive your head, gather your bones together, shake off your dust, and sit on your iron throne, so that you may eat the foreleg, devour the haunch, and partake of your rib-joints in the sky in company with the gods."[1]

These so-called Pyramid Texts, which first appear during the reign of Unas, the last ruler of the Fifth Dynasty, are the oldest written records from which the Egyptian concept of the afterlife is known. The over 700 individual spells give an incongruous, sometimes even contradictory, picture. It is not possible to say how far back in early Egyptian history, perhaps over 1,000 years, earlier ideas existed that considered reuniting limbs separated by death. At any rate, skeletons with signs of decapitation and dismemberment occurred only sporadically, and then mainly in non-elite circles. During the preparation of the dead in later eras, the restoration of an intact body, which was supposed to be produced by bandaging and embalming, among other things, was of considerable importance.

The Old Kingdom: Removal of Internal Organs and Arrangement of Outer Wrappings

The Third Dynasty (c. 2700 BC) ushered in the first peak of Egyptian civilization (Third to Sixth Dynasties, c. 2700–2200 BC): the Old Kingdom, which is characterized by great architectural monuments of stone, above all, the pyramids.

The most important evidence for a chronology of mummification techniques came from royal settings. Around 2600 BC, the removal of internal organs (evisceration) became common procedure for people, such as relatives and high officials, buried in the royal residence cemeteries. This suppressed the process of decomposition. At the beginning of the Fourth Dynasty there is only indirect proof of evisceration, in the form of niches in grave walls, each with four compartments. The compartments may have contained four organs: the stomach, intestines, lungs and liver, as this treatment was applied in later eras, when canopic jars, special containers for inner organs, or chests, were used. The earliest evidence of a canopic container was found in the tomb of Hetepheres I, mother of Khufu, whose linen-wrapped viscera were kept in an alabaster chest with four compartments. A fluid adhering to the linen proved to be a 3% natron solution—a rare early instance of the application of natron, which was not used extensively in mummification until the Middle Kingdom. The tomb of Meresanch III, granddaughter of Khufu and wife of Khafra, yields the oldest evidence for the preservation of organs in four canopic jars, which, in turn, were placed in a canopic chest. Although inner organs were occasionally removed at this time, there is still no evidence of the extraction of the brain on a regular basis during the Old Kingdom. A few individual cases perhaps indicate an experimental phase. Once removal of the inner organs became part of the process, bodies were laid out on their backs for burial.

The next step in mummification practice was typified by the mummy of a civilian, Ranefer (Fourth or Fifth Dynasty, c. 2500 BC), who was discovered in Meidum, his organs wrapped in linen and stored in a niche next to his body. Ranefer's linen-wrapped head (fig. 3) had eyes, eyebrows, hair and a mouth painted on it, and the shapes of his body parts had been reproduced, his facial features being modeled with particular care. In what had become common practice, the embalmers had completely wrapped the body in resin-soaked cloths and bandages. In addition—and also a new procedure—the abdomen had been stuffed with resin-soaked linen. This mummy, housed in the Royal College of Surgeons in London and destroyed during World War II, is a clear indication that the Egyptians of Ranefer's time attached great importance to the systematic treatment of the outer wrappings, a tradition which lasted initially until the end of the Old Kingdom, as evidenced by other mummies of this type from Saqqara, Giza and Abusir. The parts of the human body, including the genitals, were reconstructed with the aid of linen bandages and resin. Towards the end of the Old Kingdom, separate cartonnage funerary masks appeared, which, from then on were a distinguishing feature of Egyptian mummies, especially during the



3 Mummified head of Ranefer, c. 2500 BC.

Middle Kingdom and the Greco-Roman periods. Full body coverings, made of several layers of plaster with care-fully reconstructed facial features (fig. 4), which were fitted onto the bandaged contours of the body were common in the Fifth and Sixth Dynasties (c. 2500–2200 BC). Although this esthetic covering did nothing to protect against decomposition, it nevertheless restored to the body its individual shape and functionality, which were considered necessary for an existence in the afterlife.

No intact royal mummies have survived from the Old Kingdom, but there are individual body parts that can be attributed to that time, due to their discovery location and carbon-14 dating. Examples include the mummified hand of Neferefre (c. 2450 BC) and the remains of Djedkare-Isesi (c. 2400 BC), two kings of the Fifth Dynasty. The skeletal remains, including a foot wrapped in linen, which were for a long time attributed to Djoser, the most important king of the Third Dynasty, have now been radiocarbon dated to the Ptolemaic era.

Belief in the Afterworld

Towards the end of the Old Kingdom, a concept of the afterworld was established and became dominant until the end of Egyptian civilization, and even beyond. Just as the dead kings had earlier experienced the fate of the god Osiris, so did the non-royal dead as well. In mythology at first only alluded to, Osiris was the ruler of Egypt as successor to his father, Geb, and was killed by his brother and rival for the royal throne, Seth. Seth dismembered his victim and scattered his limbs throughout the land, but Osiris's sisters, Isis and Nephthys, searched for the body parts and put their brother's body back together. In the course of this archtypical mummification Osiris was not only resurrected; he also fathered a son by Isis, Horus, who avenged his father's killing and became the new ruler of Egypt. From this time on, Osiris was king and commander of the afterworld.

As a result of this, the deceased person, upon his burial, received a so-called Osiris name, which would identify him to the lord of the afterworld and thus allow his existence in the afterlife. This magical language complemented the older mummification theories but did not replace them; the meticulous preservation of bodies and the care given to the buried continued to be regarded as essential for their continued existence. Missing body parts were restored or replaced. Attempts



4 Mummy mask of a woman, painted cartonnage, Middle Kingdom, Eleventh Dynasty, c. 2050 BC, site of find unknown, probably Asyut, Roemer-Pelizaeus Museum, Hildesheim, inv. no. PM 6227.

were made to preserve the soft tissues: "O flesh of Teti, rot not, decay not, stink not." (PT §722). Any decomposition, violation or neglect of the corpse would extinguish the person once and for all. The body, therefore, had to be supplied regularly with food. This could be done by magic, through images or sacrificial texts, and the tomb had to be maintained accordingly, in order to function as the meeting place for the transfigured dead and the living.

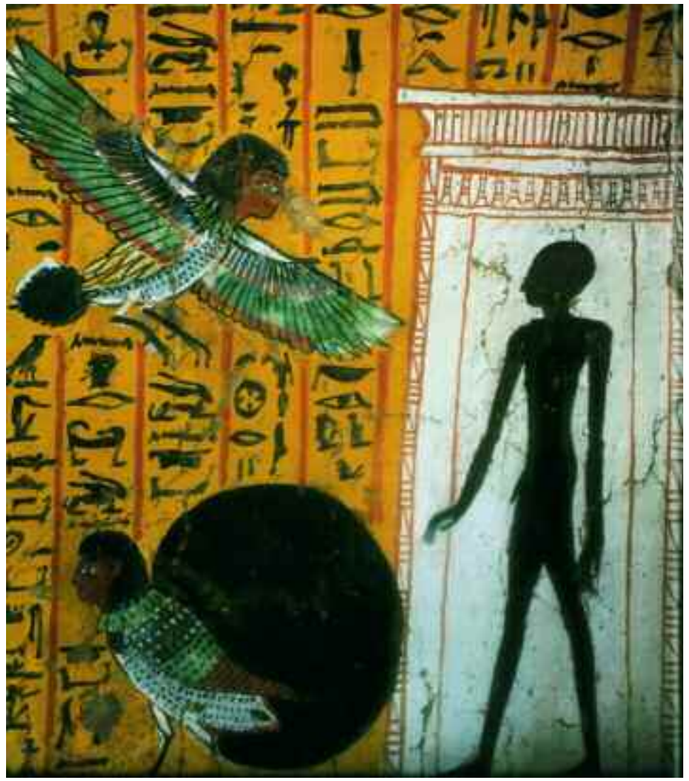
For life in the afterworld, which was supposed to be comparable to life in the earthly world, three constituents which made up a person were of great importance: *Ba*, *Ka*, and *Akh*. The *Ba* established the connection with the dead body. It became separated from the body at death, was able to fly in the form of a bird between this world and the next, and always returned to the body (fig. 5). *Ka* was the concept of a moral character, a person's double, to which social categories such as honor, dignity and status were tied. After death, *Ka* separated from the body and was able to confront the deceased. *Akh* was the spirit of the transfigured dead person. One of the aims of the funerary rites was to carry the dead person across in his *Akh*

form, to make him into a glorified ancestral spirit. The religious recitation texts needed for this were available to ordinary citizens, starting in the late Old Kingdom.

Through the help of embalming and mummification rites, a person was reconstructed, reintegrated and conveyed in a living form, so that he would be able to enter into another realm of life (heaven, underworld, or the earth's interior).

The First Intermediate Period and Middle Kingdom: A Combination of Techniques to Prevent Decomposition

The First Intermediate Period (c. 2200 BC) saw the collapse of a central authority; local rulers gained power.



5 *Ba* and shadow, from the tomb of Irinefer, west of Thebes (TT 290), c. 1250 BC.

The practice of using coffins and wrapping the body spread to provincial cemeteries, even for simpler burials. The kingdom was not reunited until about a century later, under the rule of Upper Egypt (Middle Kingdom, c. 2100–1790 BC), yet there were still autonomous local rulers who no longer had themselves buried near the king—as was still usually the case for upper officials in the previous era—but who built their own necropolises.

Mummification techniques were accordingly diverse, probably due in part to different financial means, but nonetheless a further development was observed in individual cases. Cartonnage masks, which had appeared at the end of the Sixth Dynasty, were now found on a majority of mummies. During the reign of Mentuhotep II (c. 2000 BC) this developed into the anthropoid coffin type, which housed mummies of especially high-ranking individuals. Particular importance was still given to the outer appearance of the dead body: considerable amounts of linen cloth were used for wrapping. Body parts were not only wrapped individually and bandaged together to form the typical mummy shape; the finished mummy was then sometimes wrapped in more layers of linen which formed a sort of cocoon. An example of this can be seen in the mummy of Wah from the Eleventh Dynasty, found in Thebes (fig. 6). Amulets providing protection were placed in the layers of linen. As in the Old Kingdom, the linen was sometimes partially coated with resin. Since the resin was not applied directly to the skin, however, its fungicidal and bactericidal properties did not aid the preservation process, if this, indeed, was its intended purpose.

Natron was the most important means of preserving soft tissues since the Middle Kingdom. The removal of not only of viscera, but also now of the contents of the skull, in some cases, was also helpful in preventing decomposition. This was practiced rarely and was presumably a privilege of the wealthier classes (Strouhal 1986). In the New Kingdom, from c. 1550 BC, transnasal brain removal became more common: an approximately 30 cm (11.8 in) long hooked metal rod was inserted, as a rule, through the left nostril, perforating the ethmoid bone (*lamina cribrosa*). The hook was then used to crush the brain, which was then removed through the nose. The viscera were removed, as in the Old Kingdom, through the abdominal wall; the left side was usually cut open. From the Twelfth Dynasty (c. 1990 BC) onwards, heated resin was applied directly to the skin, a treatment which could be seen in individual cases as early as in the Old Kingdom.

Mummies found in unplundered tombs are especially noteworthy. A remarkable example is Senebtisi from Lisht (Twelfth Dynasty). Her body was laid on her left side, as was typical during the Middle Kingdom, with her face pointing east. Her inner organs were removed through an incision on the left side of her abdomen; her heart, wrapped in linen, was replaced in its correct position in the body, and the abdominal cavity was then stuffed with resin-soaked cloth and sealed with resin. The brain was left in the skull. Senebitsi's mummy in its anthropoid coffin was then covered with such a large quantity of pitch that its considerable remains are still visible on the body and in the coffin (fig. 7).

None of the royal mummies from the Middle Kingdom has been preserved. The mummies of six ladies from the royal family of the Eleventh Dynasty, however, were found in shaft graves behind the temple complex of Mentuhotep-Nebhepetre in Deir el-Bahri. Rather surprising, in view of their social status, was the fact that neither the viscera nor the brain were removed. Nonetheless, they were in a good state of preservation; mummification had taken place due to the use of natron, along with resin and beeswax.

Customarily, after the organs were removed the empty abdominal cavity was filled with linen; sawdust was also sometimes used. After removal the inner organs were stored in four canopic jars, while the brain, which was regarded as unimportant (the heart being considered the seat of the mind), was discarded. The shape of the jars changed during the course of Egyptian history, but there were always four of them. From the First Intermediate Period onwards the lids sometimes featured human heads, and starting with the Middle Kingdom the containers display inscriptions putting them under the protection of the four sons of Horus: Amset, Hapi, Qebehsenuef and Duamutef. On occasion, tombs with canopic jars were found, although the internal organs were still in the body. The jars, therefore, had a significance far greater than that of being simple containers for holding the stomach, intestines, liver and lungs.

The New Kingdom: Mummification Techniques of the Royal Mummies

Regarding the rulers of the New Kingdom (c. 1550–1070 BC), who once were buried west of Thebes in the Valley of the Kings, the condition of mummy finds is extremely favorable. Although their tombs, with the well-known exception of that of Tutankhamun, were not spared from plundering, around 1000 BC caches (also known as cachettes) were set up where the defiled mummies from the Valley were reburied, in order to not to rob them of an existence in the afterlife. Two such tomb caches were discovered towards the end of the 19th century. Since the tombs of a few kings are preserved, whose mummies were missing, the existence of a third cache has been speculated. The first cache, located in Deir el-Bahri (DB 320), contained a total of 40 mummies, ten of them of pharaohs. The second cache, in the Valley of the Kings (tomb of Amenhotep II = KV 35) held 16 mummies, nine of them pharaohs (fig. 8: Kings and the royal environment). Following their reburial, the mummies remained untouched for nearly 3,000 years. Some of them are now on view in a special room in the Egyptian Museum in Cairo. Since their rediscovery, they have been scientifically investigated several times, each time using the most modern methods available. This has led to doubts as to whether the priests who inscribed royal names on some of the coffins and mummy wrappings at the time of reburial were accurate in matching the names with the mummies, since it was assumed, that some of the mummies must have been moved several times before finding their final resting place in the cachette.

The state of mummification technology during the New Kingdom can be reconstructed, thanks to the preservation of these mummies, although not every pharaoh benefited from all the various treatments. Transnasal brain removal became standard practice. Resin was applied generously, not only directly on the skin, but also poured through the nose into the brain. It was also used in the form of plugs inserted into the nasal and aural orifices, occasionally also into the anus, and served as a glue to seal the incision made into the left side of the abdomen, which was then covered by a sheet of gold. The heart always remained in the body, sometimes receiving its own linen wrappings. After removal of the organs, body cavities were cleansed with water and wine, and then stuffed with natron, resin, and/or linen. After drying, embalming and wrapping, the organs were placed in canopic jars. Particularly during the late New Kingdom, these containers sometimes had lids decorated with the characteristic heads of the four sons of Horus: Amset with a human head, Hapi with a baboon head, Duamutef with a jackal head, and Qebehsenuef with a falcon head (fig. 9). After the cleansing and drying phases, which started four days after death and lasted about 48 days, and the adornment of the outer appearance through artificial hair extensions, hair dyeing, blackened eyebrows, and other aesthetic modifications, the mummy was wrapped from head to foot, during which amulets were distributed between the linen bandages, and given its shape by stuffing its body cavities. The body was then placed in the coffin and, in some cases, then covered with a layer of resin. Embalming and wrapping took about 14 days, while seventy days were ideally needed for the entire process. The drying agents, ointments and linen cloths that were used, along with the remains of hair and nails from the embalmed body, were stored separately in containers or coffins and buried near the mummy's tomb. Whether these so-called embalming caches, some of which date back as far as the Eleventh Dynasty, were religious/ritual interments or simply rubbish pits, is currently still not clear. Towards the end of the Eighteenth Dynasty artificial eyes were first added in the form of painted pieces of linen applied over the places



6 The intact mummy of Wah (MMA 20.3.203) from Thebes (c. 2000 BC) was unwrapped at the Metropolitan Museum of Art, New York, in 1935.



7 The mummy of Senebtisi (c. 1800 BC) was so heavily covered with pitch that after the body was removed its shape could still clearly be seen in the coffin.













8 A selection of mummies from the Egyptian Museum in Cairo.

a King Seqenenre Tao II (head), died c. 1555 BC (CG 61051).

b King Seti I, front view and profile, died c. 1280 BC (CG 61077).

c King Ramses III, died c. 1150 BC (CG 61083).

d Nodjmet, consort of High Priest Herihor, died c. 1070 BC (CG 61087).

e The consort of Thutmose III or Amenhotep III (Merti or Tiye), front view and profile, died c. 1400 BC (CG 61070).

f Masaharta, High Priest of Amun (son of King Pinedjem I), died c. 1050 BC (CG 61092).

where the physical eyes were. During the Twentieth Dynasty (c. 1180—1070 BC) and continuing into the Third Intermediate Period (c. 1070—525 BC), the embalmers placed small onions under the eyelids: for example, onion bulbs were found over the compressed eyeballs of Ramses IV, the third king of the Twentieth Dynasty, whose aural and nasal orifices were sealed with resin. Since, in some cases, onions were also wrapped onto other parts of the body and were identified with the teeth of Osiris as far back as the Pyramid Texts, and in addition during the New Kingdom onions played an important role as a cult object in a feast celebrating the regeneration of the mummified Sokar-Osiris, their use was not, or not only, due to their anti-bacterial properties, as has been assumed.

In some cases the bandages were applied with particular care. The mummy of Ramses II is a good example: small animal bones were used to bandage his nose in such a way that the nostrils remained open. Great importance was given to the nose in the Egyptian imagination, since this was where the breath of life entered the body. This is why the act of smashing the nose from a statue of the deceased person was a very serious punishment, since the victim would be denied an existence in the afterlife. Incidentally, particular importance was given to the preservation of finger and toenails, which were tied to the extremities in order to prevent them from falling off during the drying process. In the New Kingdom, as a rule, any missing body parts were reconstructed and replaced during the mummification process with the aid of bandages, resin or wood. Mummification was not only a privilege of the elite in Pharaonic Egypt; persons of non-royal lineage were also embalmed for the voyage to the afterworld, albeit with less elaborate techniques: there was no evisceration, but the body was probably desiccated by the use of natron, then treated with resin and wrapped in linen bandages.

The Third Intermediate Period: The Use of Padding Under the Skin

Today, it is often stressed that the art of embalming reached its pinnacle during the Twenty-First Dynasty, around 1000 BC. At that time, many royal bodies were moved to the cachettes, which made it possible for embalmers to notice any technical faults when viewing the mummies of previous generations that had fallen victim to tomb robbers. At any rate, this period witnessed a conspicuous change in techniques. The skin was cut open on many places of the dead body, and the inner surface lined with sawdust, linen, resin or mud, in order for the body to retain its shape despite desiccation (fig. 8f). Today, the bodies which were prepared in



9 Canopic jars of Padiuf with heads of Horus's sons (Hapi = baboon, Qebehsenuef = falcon, Duamutef = jackal, Amset = human), Louvre, Paris, inv. no. N 2952.

this way offer an image of a period of renewed experimentation. In a few cases, excessive padding resulted in an unnatural appearance or in the skin bursting open, although the intention must certainly have been to create a natural image: the head and body were painted in the colors made familiar from paintings and statues—red for a man, yellow for a woman. With the advent of the Twenty-First Dynasty, burial of the inner organs outside the body fell out of use, possibly as a result of previous tomb robbing. From this point on, the organs which were removed through the side of the abdomen were placed back in the body, often together with the wax figures of the protective deities. During this time the brain continued to be removed through the nose. The skull and body cavities were filled with melted resin.

By the Twenty-Second Dynasty (c. 950 BC), the use of direct body preparation, such as shaping the body underneath the skin, declined in favor of a return to shaping the outer hull through additional linen. The preservation of soft tissue may also have been less important, although based on evidence this cannot be demonstrated convincingly. Given that there are no royal mummies preserved either from the Late Period (746–332 BC) or the Ptolemaic Dynasty (332–330 BC), it appears that the current scientifically-held view that the knowledge of mummification collapsed completely at that time is much too sweeping. Differences in the practice of mummification according to social status can clearly be recognized up to the Roman period.

The Late Period: The Greek Historian Herodotus describes the Art of Embalming

During the Twenty-Fifth Dynasty it became common to place the inner organs between the thighs or calves after their embalming; the canopic jars also found in such tombs had a purely symbolic function. The majority of Late Period mummies come from Upper Egypt and the areas bordering it (see Budde article in this volume). A tomb site in el-Assasif (Thebes) was especially rich in finds: in the funerary chamber of Wahibre a complete set of copper embalming instruments was discovered. It contained a knife, an enema device (still encrusted with brown and white matter), a spatula, tweezers, a spoon, a needle, and a hook.

As in earlier times, the preparation of the mummy depended on social status and monetary resources. Herodotus of Helicarnassus (c. 484–425 BC) wrote in his *Histories* (II, 85–8) that at the time of Persian rule there existed several methods of mummification, according to means:

> "Thus do they embalm according to the best of these ways:—First with a crooked iron tool they draw out the brain through the nostrils, extracting it partly thus and partly by pouring in drugs; and after this with a sharp stone of Ethiopia they make a cut along the side and take out the whole contents of the belly, and when they have cleared out the cavity and cleansed it with palm-wine they cleanse it again with spices pounded up: then they fill the belly with pure myrrh pounded up and with cassia and other spices except frankincense, and sew it together again. Having so done they keep it for embalming covered up in natron for seventy days [fig. 10], but for a longer time than this it is not permitted to embalm it; and when the seventy days are past, they wash the corpse and roll its whole body up in fine linen cut into bands, smearing these beneath with gum, which the Egyptians use generally instead of glue. Then the kinsfolk receive it from them and have a wooden figure made in the shape of a man, and when they have had this made they enclose the corpse, and having shut it up within, they store it then in a sepulchral chamber, setting it to stand upright against the wall.



10 Lump of natron, and other materials used during the mummification process. Collection of the Egyptological Institute Heidelberg University.

> Thus they deal with the corpses which are prepared in the most costly way; but for those who desire the middle way and wish to avoid great cost they prepare the corpse as follows:—having filled their syringes with the oil which is got from cedar-wood, with this they forthwith fill the belly of the corpse, and this they do without having either cut it open or taken out the bowels, but they inject the oil by the breech, and having stopped the drench from returning back they keep it then the appointed number of days for embalming, and on the last of the days they let the cedar oil come out from the belly, which they before put in; and it has such







11 a–c Coffin of Mut-ir-di-es from El-Hibeh, Roemer-Pelizaeus Museum, Hildesheim, inv. no. PM 1953.



12 Mummy portrait, encaustic on wood. Collection of the Egyptological Institute Heidelberg University (IN 1020, L. 36 cm, H. 25 cm) Roman, 1st century BC–1st century AD.

power that it brings out with it the bowels and interior organs of the body dissolved; and the natron dissolves the flesh, so that there is left of the corpse only the skin and the bones. When they have done this they give back the corpse at once in that condition without working upon it any more.

The third kind of embalming, by which are prepared the bodies of those who have less means, is as follows:—they cleanse out the belly with a purge and then keep the body for embalming during the seventy days, and at once after that they give it back to the bringers to carry away."[2]

There is archaeological evidence for the second and third types of embalming, while it can be assumed that tomb robbers destroyed the mummies of the first, more costly type during looting.

The Embalming Ritual

Egyptian text sources do not describe the methods used for cleaning and drying the corpse. The techniques of wrapping and embalming, on the other hand, are found in the so-called embalming ritual, which is preserved fragmentarily in two papyri from the 1st century AD. The following text is one of the eleven paragraphs, each consisting of technical instructions and a collection of spells to be recited (Boulaq III papyrus, 7.1–7.7):

"After this, anoint his head with oil of myrrh. Perform this—i.e. anoint the head of the transfigured one with oil of myrrh—a second time, and this time with the oils named "binding the head" and "binding the face". Under his head apply a layer of adhesive consisting of powdered myrrh and oil of juniper.

Under his head the following is then recited: O you, Osiris- [here the name of the deceased is inserted]: receive your head in the west, that you may enter among the ranks of the blissful transfigured. Your tomb has been appointed flawlessly, and it is provisioned exactly for you in the necropolis. Your name is illustrious in the workshop of the embalmers, for you praise the blissful transfigured. The dwellers of Duat [the underworld] kiss the earth before your corpse; the dwellers of Heaven receive your *Ba*, the dwellers of Earth give you praise. The dwellers of the valley cleanse your corpse, and Anubis and Horus render your embalming most perfectly. Thoth makes your limbs sound through his utterances. O you, Osiris, Father of the Gods- [here again the name of the deceased is inserted]: you are vindicated before the great royal tribunal, which is in Heliopolis, before the great Gods of the Temple of Re, before the great royal tribunal, which is in the temple of Ptah. Your head will come to you and not part from you. It will most surely come to you and not part from you, eternally!"[3]

In such ritual texts, which were written 2,500 years after the Pyramid Texts, the deceased was still identified with Osiris, whose fate he undergoes in order to be able to continue living in eternity. The embalming ritual depicted the corpse as a body that was not only lifeless, but, most importantly, dismembered. This state of dismemberment was overcome physically through bandaging; medically and medicinally through surgical intervention and mummification agents—which, incidentally, are referred to as "drugs"—and also magically, through the spells which accompany each procedure. In the end, the mummy is created in the image of the god Osiris. Along with mummification, other rites were considered necessary for the transformation, resurrection, and glorification of the deceased, among them funerary processions, the opening of the mouth ritual, and protective rites at the tomb. A small visual depiction of the body's treatment has been preserved on the coffin of Mutirdis (fig. 11) from El-Hibeh, dating from the 2nd century BC. It shows, among other things, the washing and purification of the body, and afterwards its laying-out. One of the priests present in the embalming hall is wearing an Anubis mask, and offerings and a boat bearing Osiris, god of the dead, can also be seen.

13 Mummy with a portrait of a young man, tempera painting on linen, 20 AD, from Hawara, Egyptian Museum of the University of Leipzig, inv. no. 1683.



The Greco-Roman Period: Individualized Mummies and Refined Bandaging Technique

The bodies embalmed for the afterlife in Ptolemaic and Roman Egypt (332 BC–395 AD) today present a disparate picture of varying techniques and local customs, not only in their outward appearence, but also as far as organ preservation is concerned. Great importance continued to be given to the outer appearance of mummies. A large number of variations were noted from among the mummies found in the "Valley of the Golden Mummies". This necropolis, which was discovered quite recently, in 1996, in the Bahariya Oasis, holds 105 intact mummies just in the area excavated so far. They date back to the first century AD. The bodies of the elite were completely and meticulously bandaged with strips of linen, and their heads and chests were covered by cartonnage masks which were painted and, on 60 of the masks, gilded. The gods on the breastplates and the depictions of scenes from the Egyptian Book of the Dead show that the old beliefs about the afterworld were still widespread, although the masks, with their contemporary hairstyles, headgear and facial expressions, were fashioned in the Roman style.

It was probably primarily members of the middle class who were bandaged in linen, but had to do without masks, painting and gold. The bandaging was done extremely carefully to produce a geometric pattern (criss-cross pattern). By contrast, the poorest Oasis inhabitants were rather carelessly bandaged, so that in a few cases the bandages have fallen apart over time. Extensive analysis of the preservation methods in the Bahariya mummies have yet to be carried out, but data from other excavations provide some relevant information. The anthropological investigation of bodies from the Douch necropolis (end of the Ptolemaic era until the mid-5th century AD) in the Kharga Oasis has revealed that various methods were employed to prevent decomposition, apparently depending on the status of the deceased. In the most elaborate type of burial, which can be distinguished superficially by the traces of gold on the face, hands, and feet, the brain and abdomen were emptied and the skull was filled with embalming material. In a second group of mummies, the brains were missing, although the bodies contained all of their viscera; the skin on the bodies was anointed with embalming agents. The mummies of a third type were not chemically treated and the interior of the



14 Female mummy from Antinopolis from the 4th century AD, natural mummification. Burial from the Roman period. Royal Museum of Art and History, Brussels, inv. no. 1045.

bodies was left intact, which led to an advanced state of decomposition.

The human remains from the necropolis at Kellis (end of Ptolemaic era to 400 AD) in the Dakhla Oasis can also be differentiated according to their mummification status: some of the bodies are natural mummies, while others have been preserved through human intervention. In total, 94% of these bodies had their brains removed transnasally; in contrast to the Kharga mummies, however, no embalming liquids had been poured into the skull. In some of the deceased, the inner organs were removed through an incision in the left side of the body or through the rectum, and the body cavities were filled with embalming matter and resin-soaked balls of linen. In addition, hot resin had been applied to the skin—in some cases, this was the only action taken. There is no evidence that natron had been used to dry out the corpse.

It is also known from tombs in the Nile Valley that the methods used often varied. Evisceration was encountered less in mummies from the Greco-Roman period, and the use of natron seems to have ceased completely. All the more conspicuous is the generous use of embalming agents, which were applied in the body's cavities as well as on its surface. In those cases where the brain had been removed, this was done either in the traditional way, through the ethmoid bone (*lamina cribrosa*) and the nasal cavity, or in a simpler way, through the large hole at the bottom of the skull (*foramen magnum*).

Certain visible features are typical in mummies of the Greco-Roman period. First, a refinement of bandaging techniques was noted (fig. 13). Plaster masks on mummies, mainly from Hermopolis in Middle Egypt, have been preserved. A distinctive feature that originated in Egypt during Roman domination (1st–3rd centuries AD) are the so-called portrait mummies, a large number of which have been found in the necropolises of the Faiyum Oasis, Antinopolis, and Akhmim. The mummies were carefully wrapped and their faces are overlaid with a thin wooden board bearing a portrait of the face of the deceased painted in encaustic or tempera technique (figs. 12, 13). The oldest known mummies of this type come from the reign of the emperor Tiberius (14–37 AD), the most recent from the 4th century AD. The main motive for the introduction of portraits into the cult of the dead is considered to be the ever-increasing desire for overt self-portrayal. The deceased was no longer idealized, but personalized, although he continued to identify himself with Osiris. Members of the Roman elite living in Egypt were able to adopt Egyptian concepts of the afterlife, as proven by images of funerary art.

There is also a visible change to burial practices. Sometimes, apparently, years could pass between death and burial, during which time the mummified person would remain in the house of his or her surviving family. The Greek-writing satirist Lucian of Samosata (c. 120–after 180 AD), who spent several years in Egypt, described it (*De luctu* 21): "the Egyptian

embalms … In Egypt, indeed, the corpse, duly dried, is actually placed at table—I have seen it done."[4] And Saint Anthony, an Egyptian monk, was afraid he might not receive the burial due to a pious Christian, according to Athanasius, bishop of Alexandria (c. 298–373): "if you have any care for me and are mindful of me as of a father, suffer no one to take my body into Egypt, lest haply they place me in the houses."[5]

The End of Mummification

With the Edict of Theodosius in 392 AD, all "heathen" cults were prohibited. The practice of mummification, however, did not fall under the ban, since Coptic Christians continued occasionally to preserve the bodies of especially venerated persons by the use of salt (without, however, evisceration), then storing and taking care of them (fig. 14). In Christianity, and related cultures, the corpse as a companion of the living has never become obsolete, as exemplified by Lenin's mummy (whose brain was removed for "scientific" investigation) and the Capuchin catacombs of Palermo,; as well as the highly complex Christian cult of relics.

In Egypt, meanwhile, the expansion of Islam in 641 AD put an abrupt end to the religious importance of the human corpse. According to Muslim rite, if a person dies in the morning he is buried on the same day, otherwise burial is on the next day. The body—unpreserved, but washed and wrapped in white cloth—is given to the earth of Europe, which destroys, or to the sand of the desert, which still desiccates; respect for the peace of the deceased is so great that almost any type of ornament or maintenance for the grave is dispensed with, at least in the case of ordinary mortals.

Appendix: The Most Important Embalming Agents

Evidence for the agents used in embalming is found in ancient texts as well as in the material remaining on mummies and vessels. Analyzing the text sources, such as from the above-quoted embalming ritual, the ritual book for the embalming of the Apis bulls, the accounts from embalming workshops, and the inscriptions from vessels found in embalming caches, is problematic, since the ingredients that are named cannot all be identified with certainty. Only the substances natron, asphalt, wax, pitch, and myrrh can be safely translated from their historic meaning.

Natron was used as a desiccant in mummification. Chemically speaking, this was not lye—which some modern translations of the word borrowed by the Greeks and used in Herodotus would have us think—but a mixture of sodium carbonate and sodium bicarbonate, together with smaller amounts of sodium chloride and sodium sulfate, such as was collected by the ancient Egyptians, mainly on the shores of Wadi el Natrun Lake. The desiccating effect comes mainly from the crystallized sodium carbonate, to which up to 10% water can bind without appearing wet. Dehydration could be hastened by repeatedly replacing the layers of salt; wood shavings and linen also supported the process.

After dehydration the body was treated with ointments and resins, sometimes also bitumen. An Egyptian word for "bitumen" was not encountered until the New Kingdom, but the substance was probably already known through trade relations with Mesopotamia. There, it was employed to seal structures against water, a use the Egyptians noticed and adapted for the purpose of embalming corpses. The relatively late appearance (during the Twentieth Dynasty) of an Egyptian word for bitumen is matched by archaeological finds. Analysis of mummies shows the first evidence of bitumen in the 9th century BC (Twenty-Second Dynasty); at that time it was extracted from natural deposits at Gebel Zeit (Mons Petrolius) and Abu Durba on the shores of the Gulf of Suez. In the Greco-Roman period mummies were also treated with bitumen from the Dead Sea, which is confirmed not only by analysis, but by Greek writings. Diodorus Siculus (19.9) mentions the bitumen of the Dead Sea thus: "The barbarians who enjoy this source of income take the asphalt to Egypt and sell it for the embalming of the dead; for unless this is mixed with the other aromatic ingredients, the preservation of the bodies cannot be permanent."[6] Posidonios similarly remarks in Strabo (16.2): "The Egyptians use the asphalt for the embalming of their dead."[7] Like resin and beeswax, the black-brown substance softened when heated and was used alone or in combination with other salve ingredients—there is evidence of blending with animal and vegetable oils. Due to its liquid-repellant properties, bitumen protected natron-dried bodies from rehydration and subsequent decomposition.

Beeswax had a similar effect and was used in mummification from the Middle Kingdom. Beeswax was not mentioned as an embalming agent in medieval or early modern texts, but in Egyptian embalming texts



15 Shroud from Saqqara, 2nd century AD, Trier, Research Center for Greco-Roman Egypt, inv. no. OL 1997.11.

Korrektur unklar

Korrektur unklar

it appeared exactly in those places where bitumen was not named.

Resins were already applied to linen in 4000 BC, although the antiseptic properties were effective only when they were in direct contact with the skin. The remains of resin in mummies were mainly from conifers, in particular the Aleppo pine (*Pinus halepensis Mill.*). Since Egypt was poor in resin-rich tree species, the embalming agent had to be imported. Starting in the Eighteenth Dynasty, resin from cedars (*Cedrus atlantica (Endl.) Manetti*) was used in mummification. Depending on the particular type of resin available from traders at the time, whether resinous wood or lumps of dried resin, various techniques were needed to prepare the embalming substance. A viscous, spreadable preparation was obtained by warming the resin, but heating it a high temperature under airtight conditions created pitch. Some texts actually refer to it as a "hot drug". Shrouds from the Roman period also feature the jackal-headed Anubis, the god of the dead responsible for embalming, in his role as preparer of ointments, cooking the mummification substance on a stove (fig. 15). Resin was prepared in large amounts, particularly during the Roman era. The chemical transformation processes induced by high temperatures in the wood tar samples secured by archaeologists can be followed analytically through their typical ingredients. The antibacterial and antifungal properties of wood tars have long been known, but a more recent discovery is that some of the components they contain, principally guaiacol and other phenols, have such potent preservative qualities that an alkaline bone enzyme found in Egyptian mummies is still immunologically and enzymatically active, even today.

Both Herodotus and Diodorus mention "kedros" as an embalming agent. Pliny (XXIV, [21] 52), writing about *kedros*, says that it was the name for liquid pitch in Syria, and that it in Europe it is prepared from the Arolla pine. He described a method of extracting pitch from wood which is rich in resin. For a long time it was debated as to exactly which resin *kedros* referred; as a rule, the *kedros* of Herodotus and Diodorus is translated as "cedar resin". Since *kedros* was a word used by the Greeks for several plant species which are not explicitly differentiated, namely, for instance, cedar species, such as *Cedrus libani A. Rich.*, and also species of juniper, such as *Juniperus oxycedrus L.*, the question arose as to whether it was cedar resin or oil of juniper (juniper species do not yield resin, but instead an essential oil) that was used as an embalming agent. Analysis speaks convincingly in favor of cedar, however; it is important to note that traces of Aleppo pine resin have also been found in mummies dating back as far as the Twenty-Third Dynasty, as has pistachio resin from *Pistacia lentiscus L.*

Among the embalming agents mentioned by Herodotus and Diodorus was a substance that, so far, has not been found through analysis: cinnamon. The fact that it was named as a scent, plus the lack of any surviving traces of it, indicate that cinnamon was not used as a solid substance, but rather as a scented oil, whose delicate components have long since evaporated. Cinnamon—the cassia of which Herodotus spoke, is now thought to be "China" cinnamon, while the *cinnamomum* mentioned by Diodorus is "Ceylon" cinnamon. It was apparently a very unusual ingredient, originating as it did in faraway regions. There is, at present, no archaeological evidence for its importation into Egypt.

Myrrh also had to be imported, and was also counted among the most precious of ingredients. Today, the source plant of myrrh for pharmaceutical uses is considered to be *Commiphora molmol* (Engl.), but the resin from other species of *Commiphora* (family *Burseraceae*) is also referred to as myrrh. This liquid gum resin, which seeps out of cracks in the tree bark and hardens on contact with air, gives off a faint, aromatic odor and has a lingering bitter taste. In the countries where it is collected it is separated into different grades of quality, according to its composition, so that imported resin may consist of a mixture of resin coming from different source plants. The genus *Commiphora* comprises over two hundred species and grows from the Red Sea to South Africa, and in India. Just as we can hardly determine the origin of individual resins today, so must we also accept that the term "myrrh resin" in antiquity has a very wide-ranging meaning.